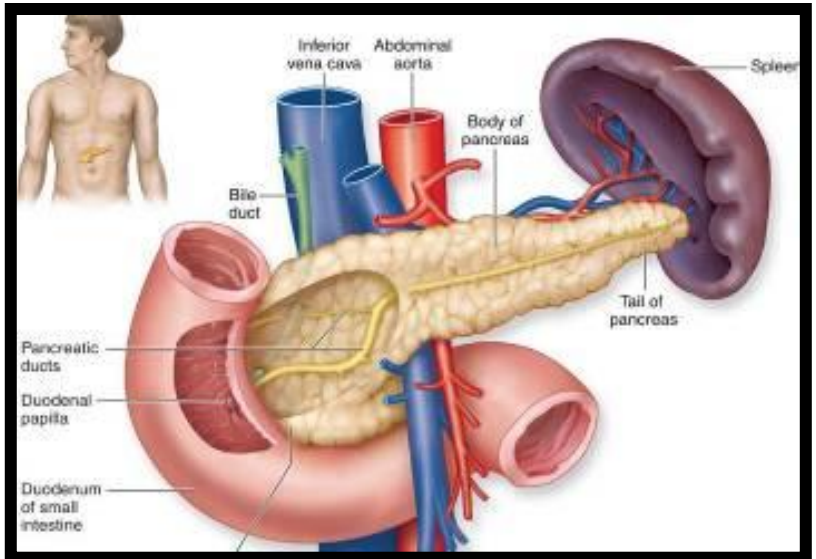
MICROSCOPIC STRUCTURE OF PANCREAS

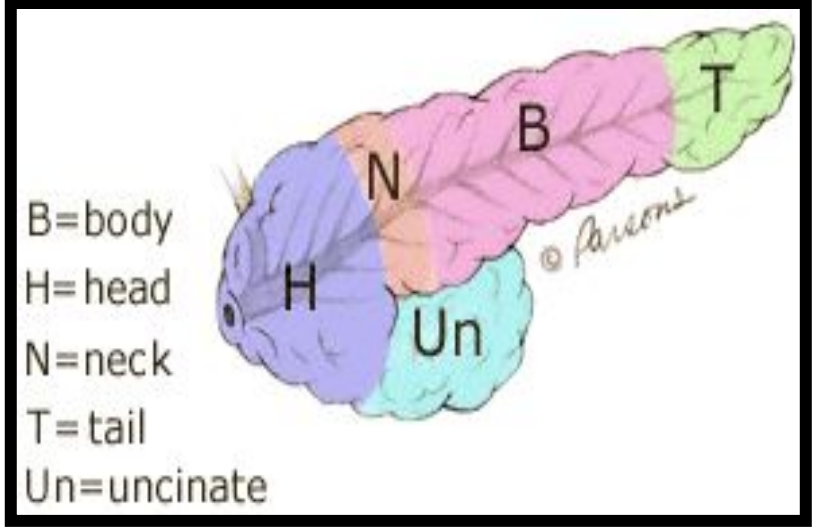
LEARNING OBJECTIVES

- Brief review of gross anatomy of pancreas
- Discuss the histological components of pancreas
- Describe the capsule and stroma pancreas
- Discuss the Parenchyma and Lobules (acini) of Pancreas
- Discuss the Duct System of Pancreas
- Describe the endocrine component of pancreas
- Discuss the differences between Parotid gland and Pancreas

GROSS FEATURES OF PANCREAS

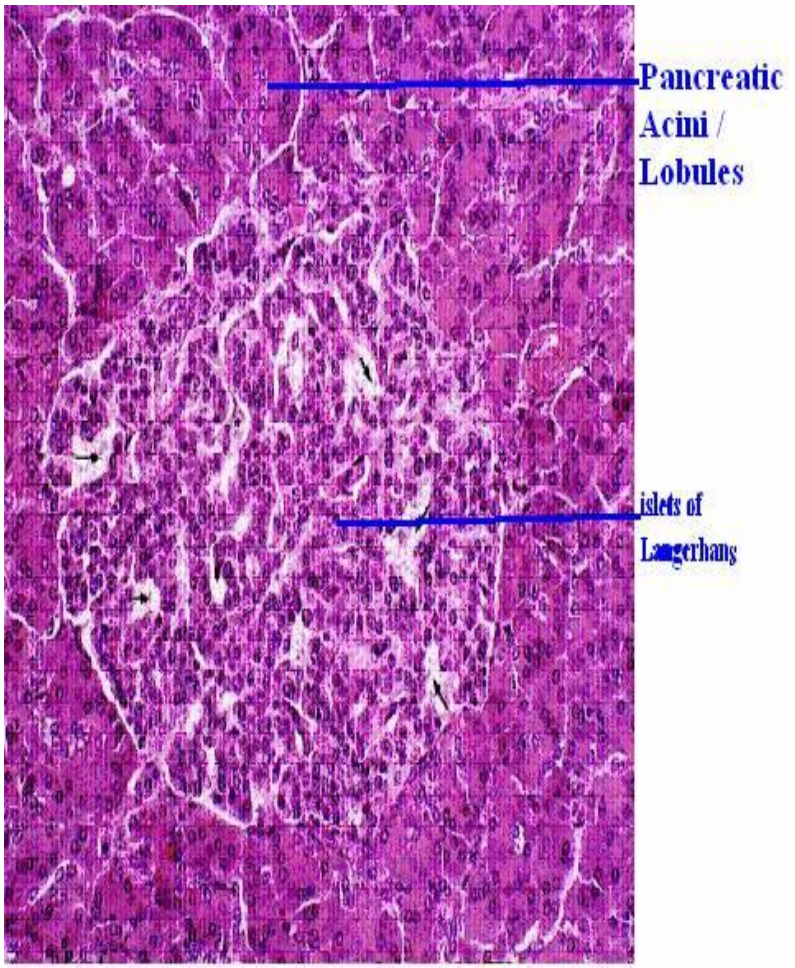
- It lies in the curvature of duodenum.
- The pancreas lies mostly posterior to the stomach.
- It opens in the 2nd part of duodenum by the Main Pancreatic duct.
- It extends across the posterior abdominal wall from the duodenum, on the right, to the spleen, on the left.





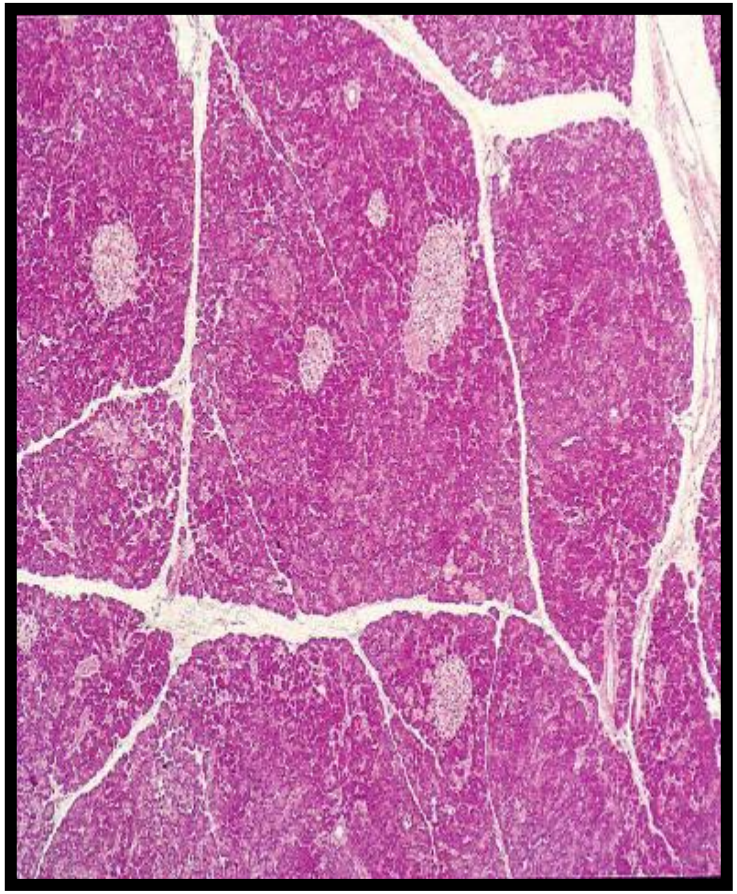
HISTOLOGICAL COMPONENTS OF PANCREAS

- The pancreas is a mixed exocrine and endocrine gland that produces digestive enzymes and hormones.
- The enzymes are stored and released by cells of the exocrine portion, arranged in **lobules or acini.**
- The hormones are synthesized in clusters of endocrine epithelial cells known as **islets of Langerhans**



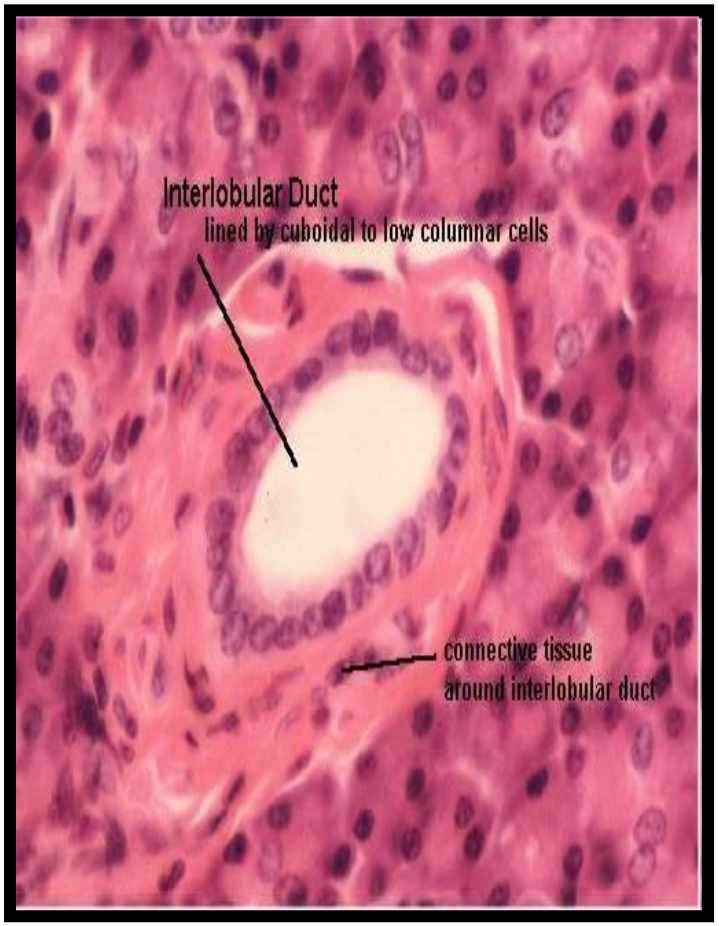
SECRETORY ACINUS

Pancreatic exocrine cells are arranged in grape-like clusters called ACINI (a single one is an ACINUS). The exocrine cells themselves are packed with membrane-bound secretory granules which contain digestive enzymes that are exocytosed into the lumen of the acinus.



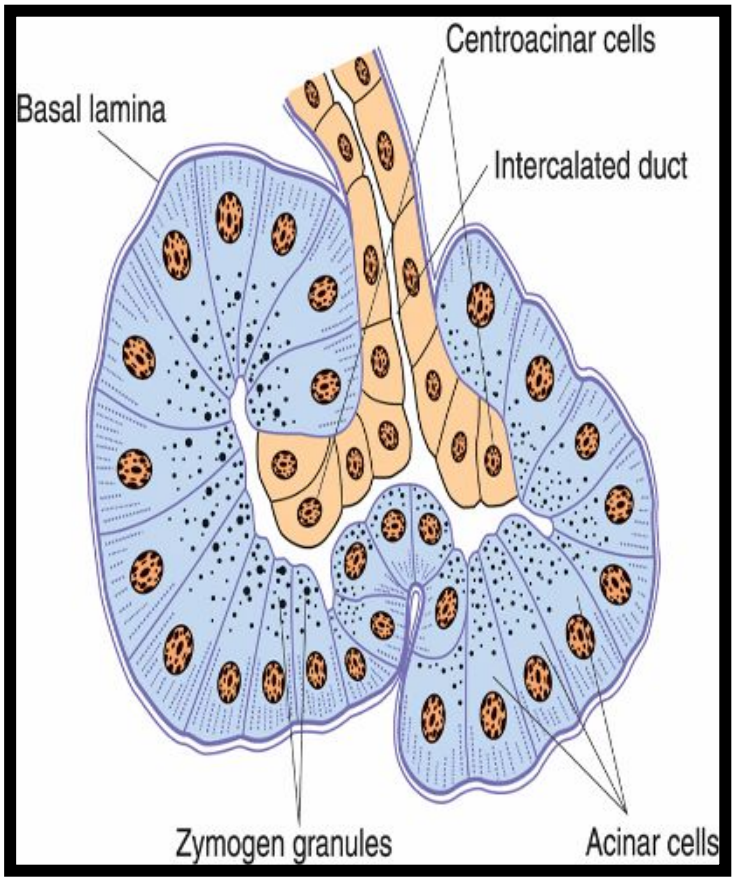
CAPSULE AND CONNECTIVE TISSUE

- The pancreas consists of two components the parenchyma (secretory part) and stroma (Connective tissue part).
- A thin connective tissue capsule covers the pancreas and sends septa into it, separating the pancreatic lobules.
- The acini are surrounded by a basal lamina that is supported by a delicate sheath of reticular fibers.
- The pancreas also has a rich capillary network, essential for the secretory process.



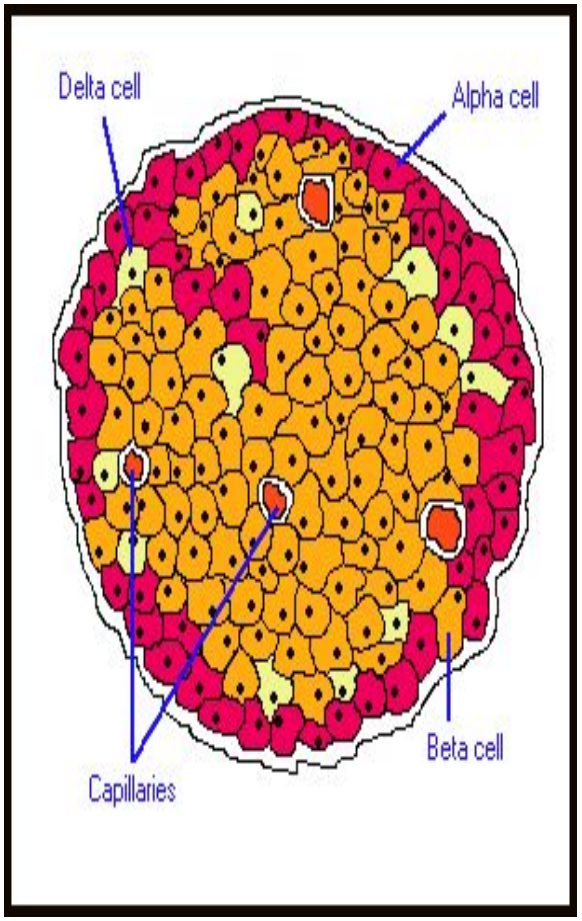
CENTROACINAR CELLS

- The centers of the acini frequently contain one or more nuclei of **Centroacinar Cells** with pale nuclei and sparse pale-stained cytoplasm;
- These represent the terminal lining cells of intercalated ducts.

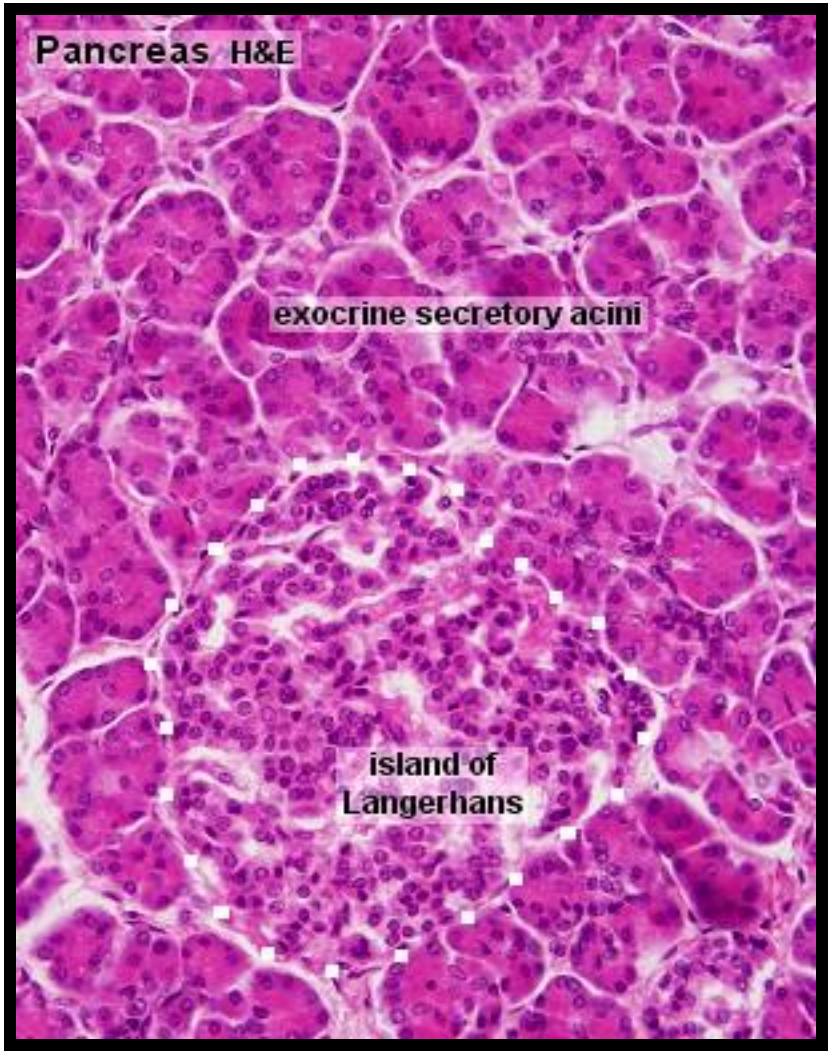


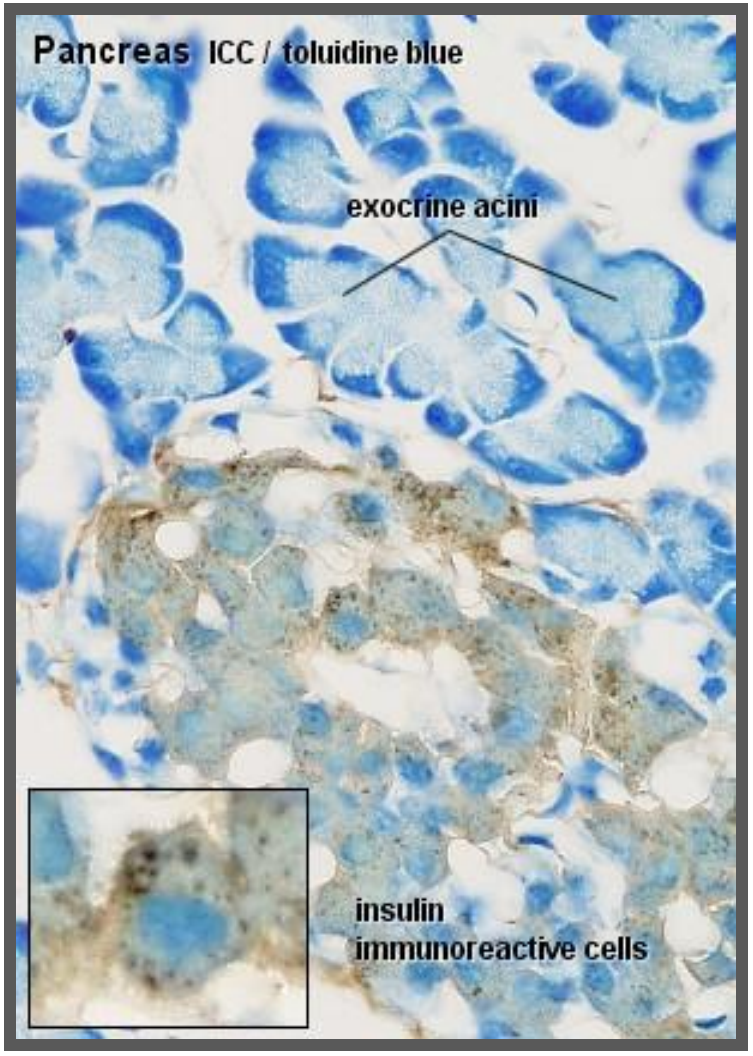
ENDOCRINE PORTION OF PANCREAS

- Embryonic epithelium of the pancreatic ducts consists of both potential exocrine and endocrine cells.
- During development, the endocrine cells migrate from the duct system and aggregate around capillaries to form isolated clusters of cells, known as **islets of Langerhans**, scattered throughout the exocrine glandular tissue.
- The islets vary in size and are most numerous in the tail of the pancreas.
- The islets contain a variety of cell types each responsible for secretion of one type of polypeptide hormone.



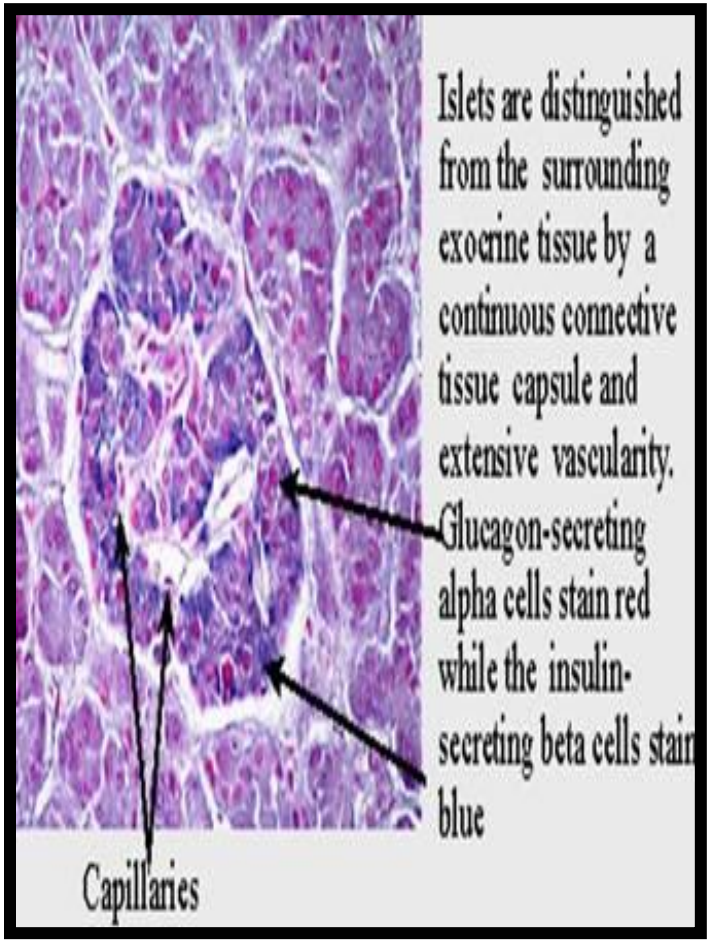
ISLETS OF LANGERHANS





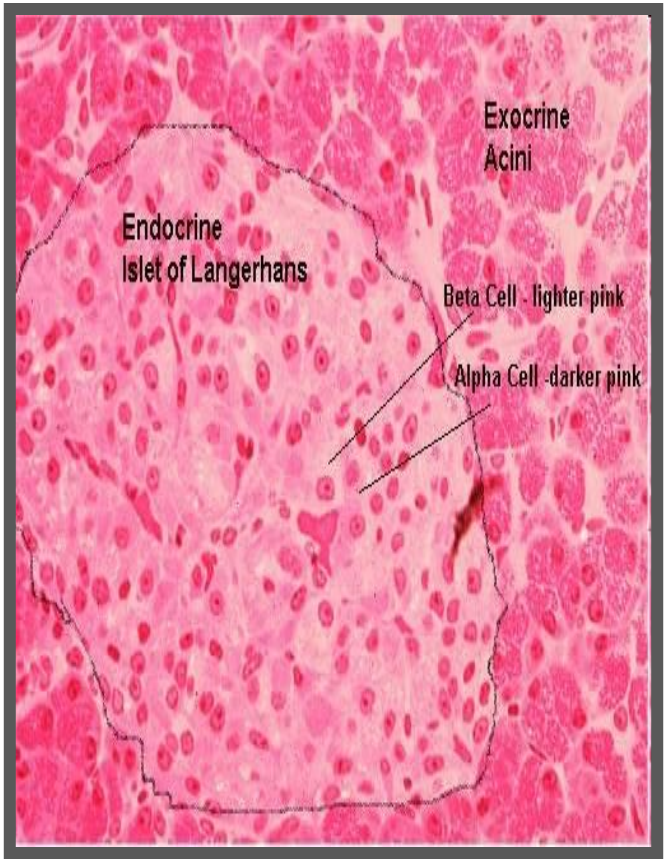
ISLETS OF LANGERHANS

- The islets of Langerhans are multi-hormonal endocrine micro-organs; they appear as rounded clusters of cells embedded within the exocrine pancreatic tissue.
- Most islets are 100–200 μm in diameter and contain several hundred cells, small groups of endocrine cells are also found interspersed among the pancreatic exocrine cells.
- Here may be more than 1 million islets in the human pancreas.
- A fine capsule of reticular fibers surrounds each islet, separating it from the adjacent pancreatic tissue.



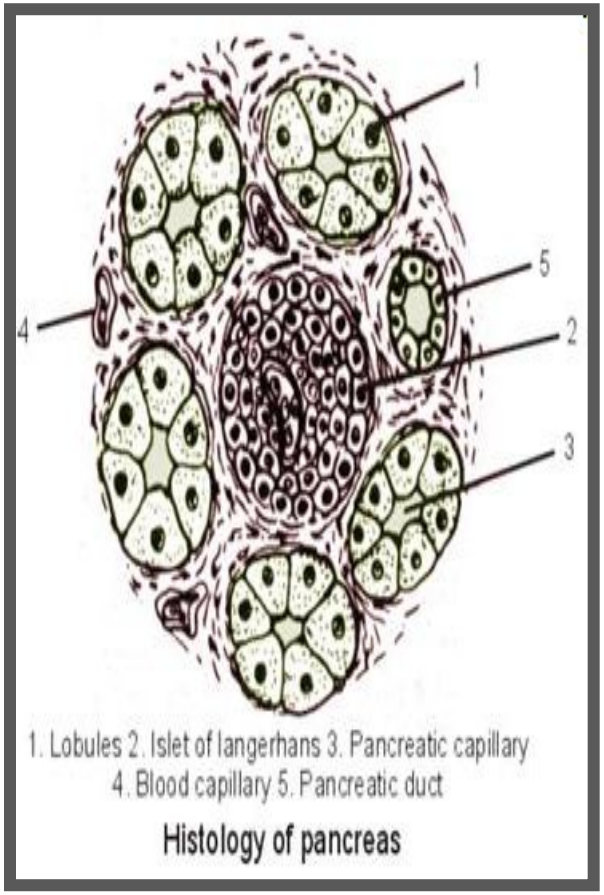
ISLETS OF LANGERHANS

- Each islet consists of lightly stained polygonal or rounded cells, arranged in cords separated by a network of blood capillaries.
- Trichrome stains allow the recognition of acidophils (α) and basophils (β).
- While using immunocytochemical methods, four types of cells—A, B,D, and F—have been recognized in the islets.



ISLETS OF LANGERHANS

- The ultrastructure of these cells resembles that of cells synthesizing polypeptides.
- The secretory granules of cells of the islets vary according to the species studied.
- In humans, the A cells have regular granules with a dense core surrounded by a clear region bounded by a membrane.
- The B (insulin-producing) cells have irregular granules with a core formed of irregular crystals of insulin in complex with zinc.



1. Lobules 2. Islet of langerhans 3. Pancreatic capillary
4. Blood capillary 5. Pancreatic duct

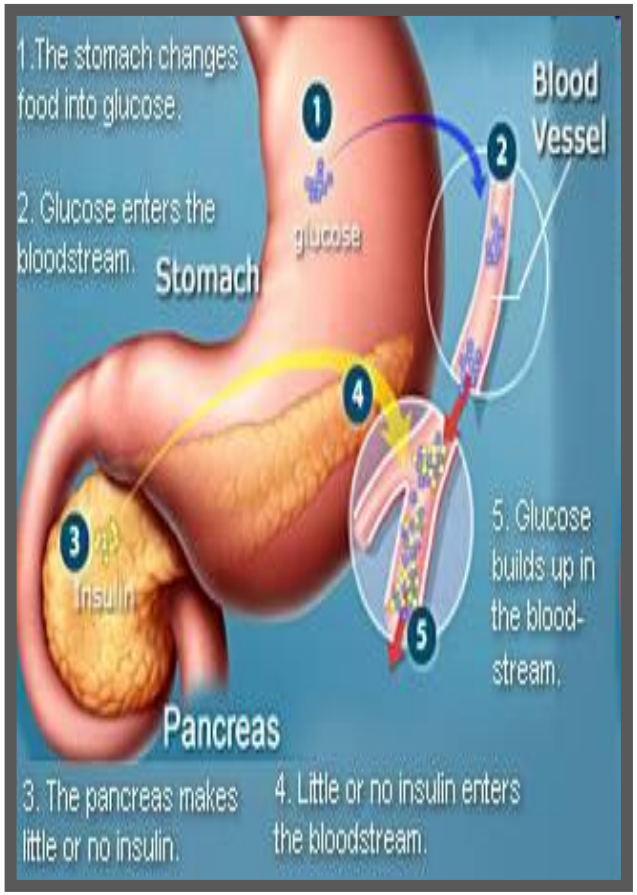
Histology of pancreas

The relative quantities of the four cell types found in islets are not uniform; they vary considerably with the islet's location in the pancreas as shown in table.

Cell Type	Quantity	Position	Hormone Produced
A	20%	Usually in periphery	Glucagon
B	70%	Central region	Insulin
D	<5%	Variable	Somatostatin
F	Rare	Variable	Pancreatic polypeptide

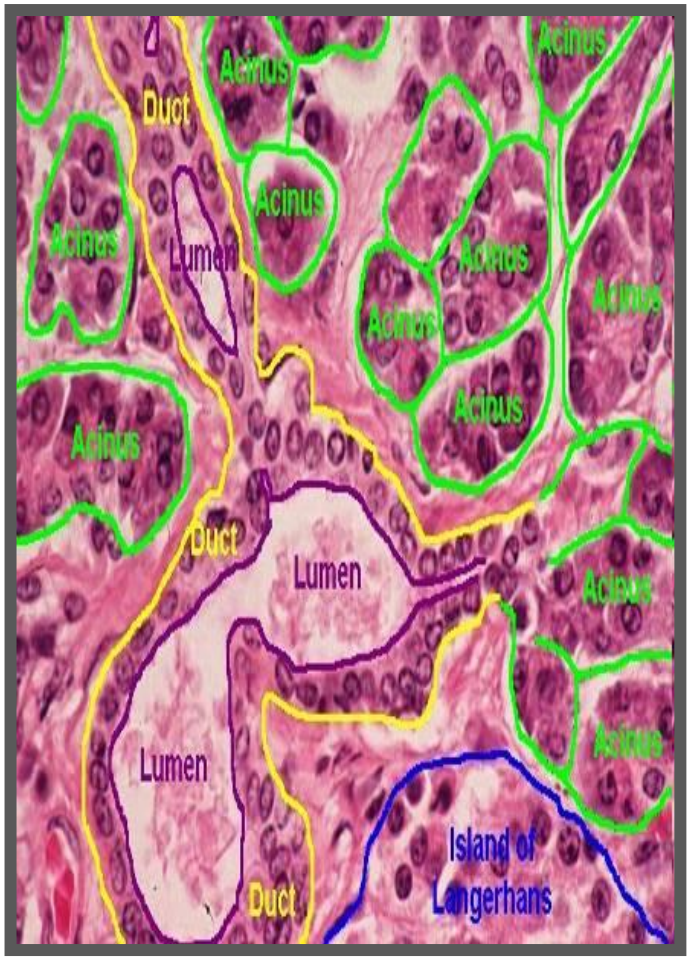
NERVE SUPPLY OF ISLETS OF LANGERHANS

- Both the endocrine cells and the blood vessels of the islets are innervated by autonomic nerve fibers.
- Sympathetic and parasympathetic nerve endings have been found in close association with about 10% of the A, B, and D cells.
- These nerves function as part of the insulin and glucagon control system.



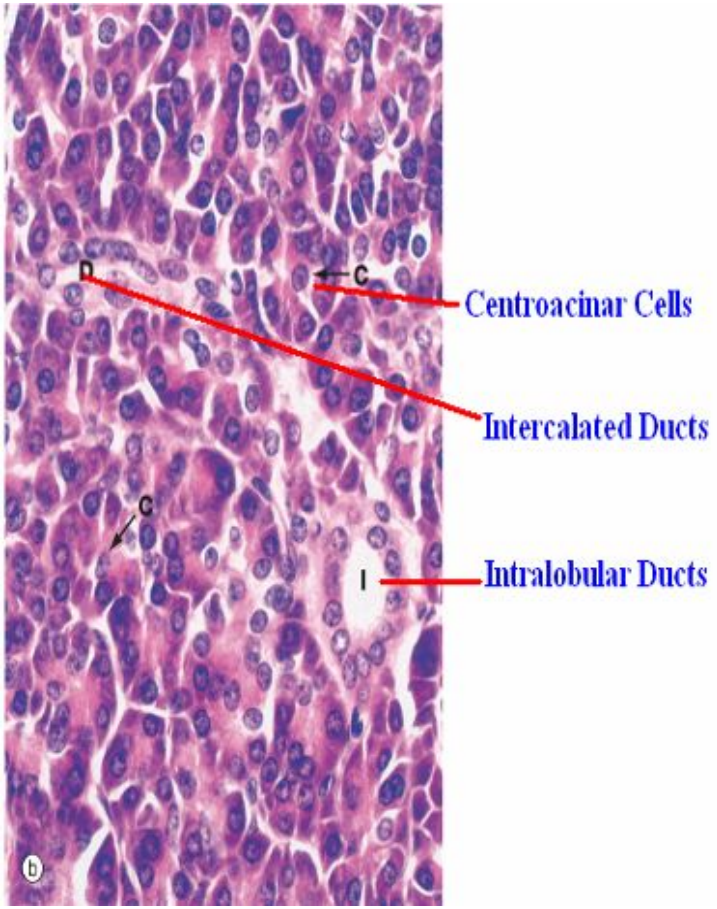
DIABETES MELLITUS

- Insulin-dependent or type I diabetes (juvenile diabetes) results from partial or total destruction of B cells due to an autoimmune disease, the individual susceptibility to which is controlled by several genes.
- Insulin-independent diabetes or type II diabetes occurs at a later stage in life and is frequently associated with obesity.



DUCT SYSTEM OF PANCREAS

- The **intercalated ducts** drain into small ***intralobular ducts***, which in turn drain into the ***interlobular ducts*** in the septa of the gland.
- The intercalated ducts are lined by simple low cuboidal epithelium, which becomes stratified cuboidal in the larger ducts.
- With increasing size, the ducts are invested by a progressively thicker layer of dense collagenous supporting tissue; the wall of the main pancreatic duct contains smooth muscle.



DUCTS OF PANCREAS

Duct System of Pancreas

↓

Intercalated Ducts

↓

Intralobular Ducts

↓

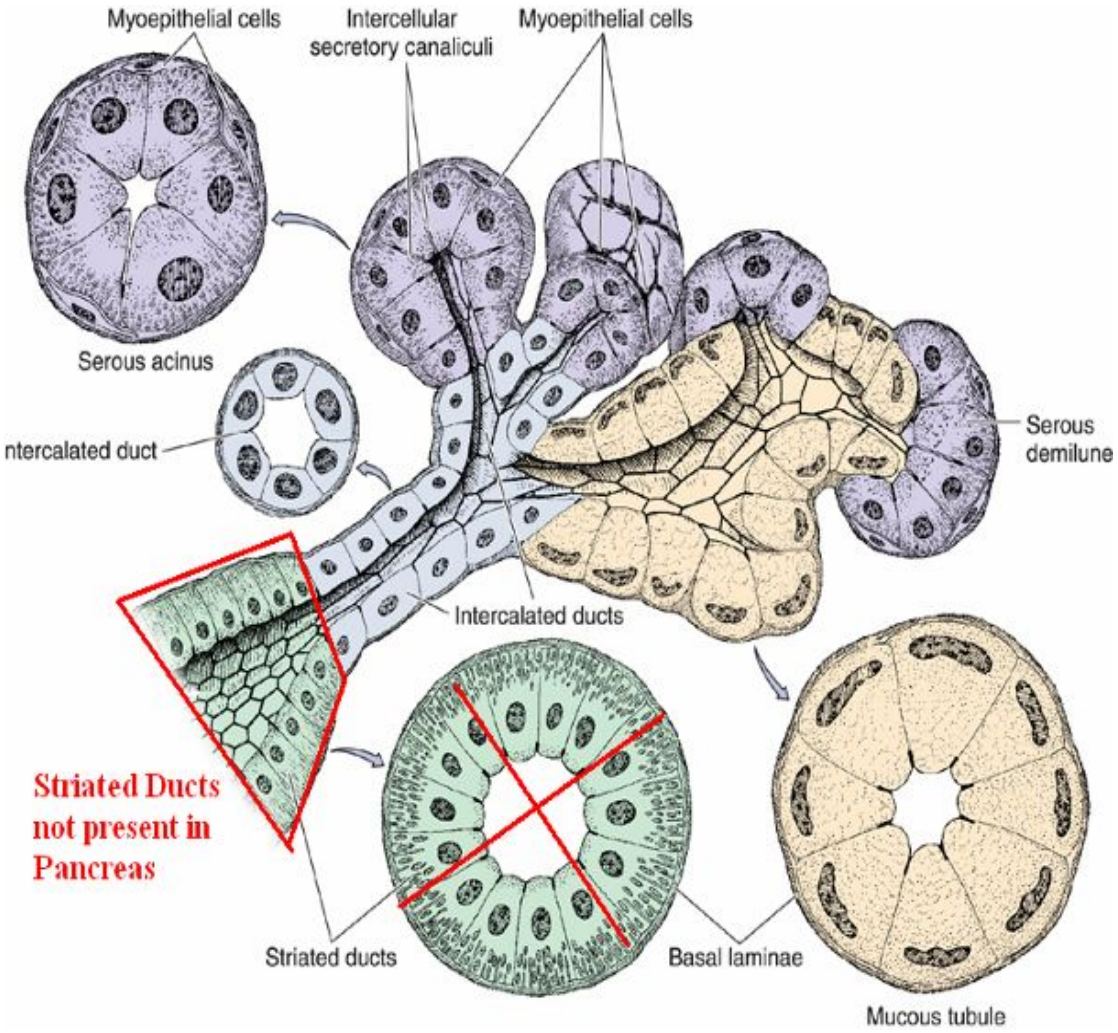
Interlobular Duct

↓

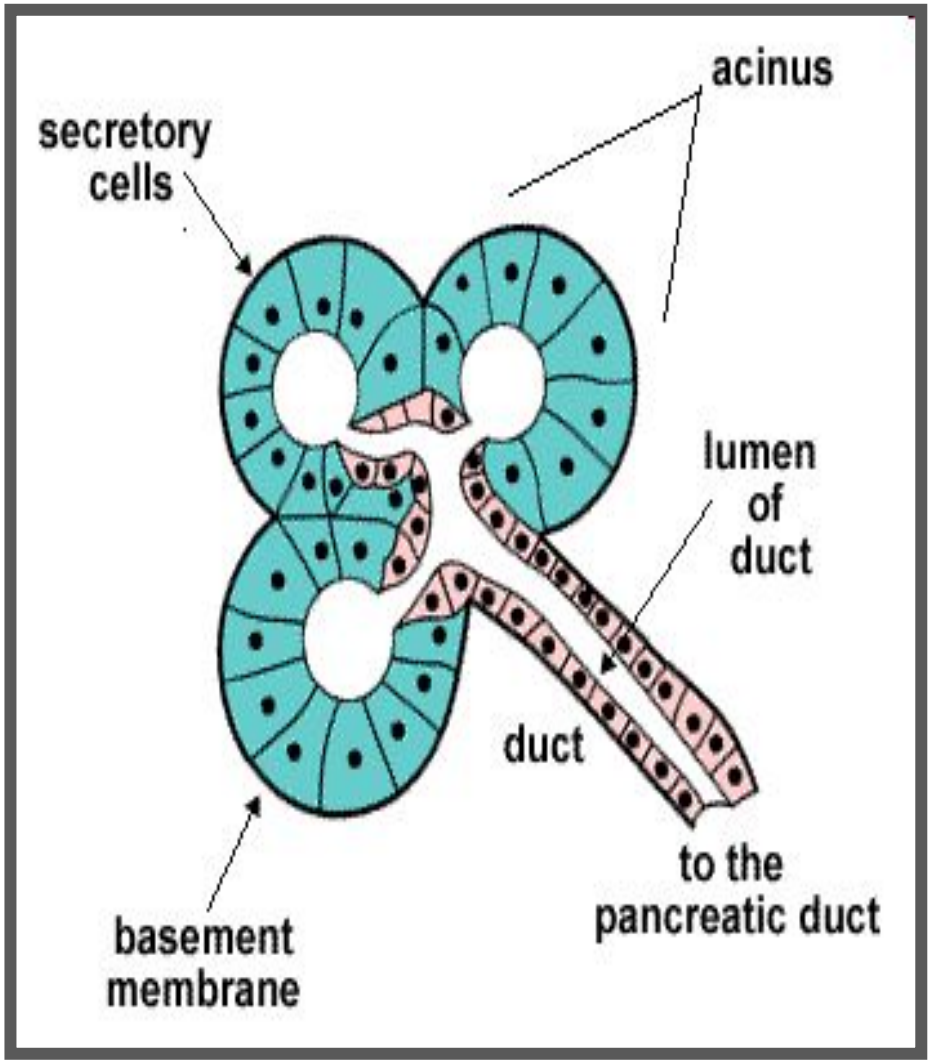
Larger Excretory Duct

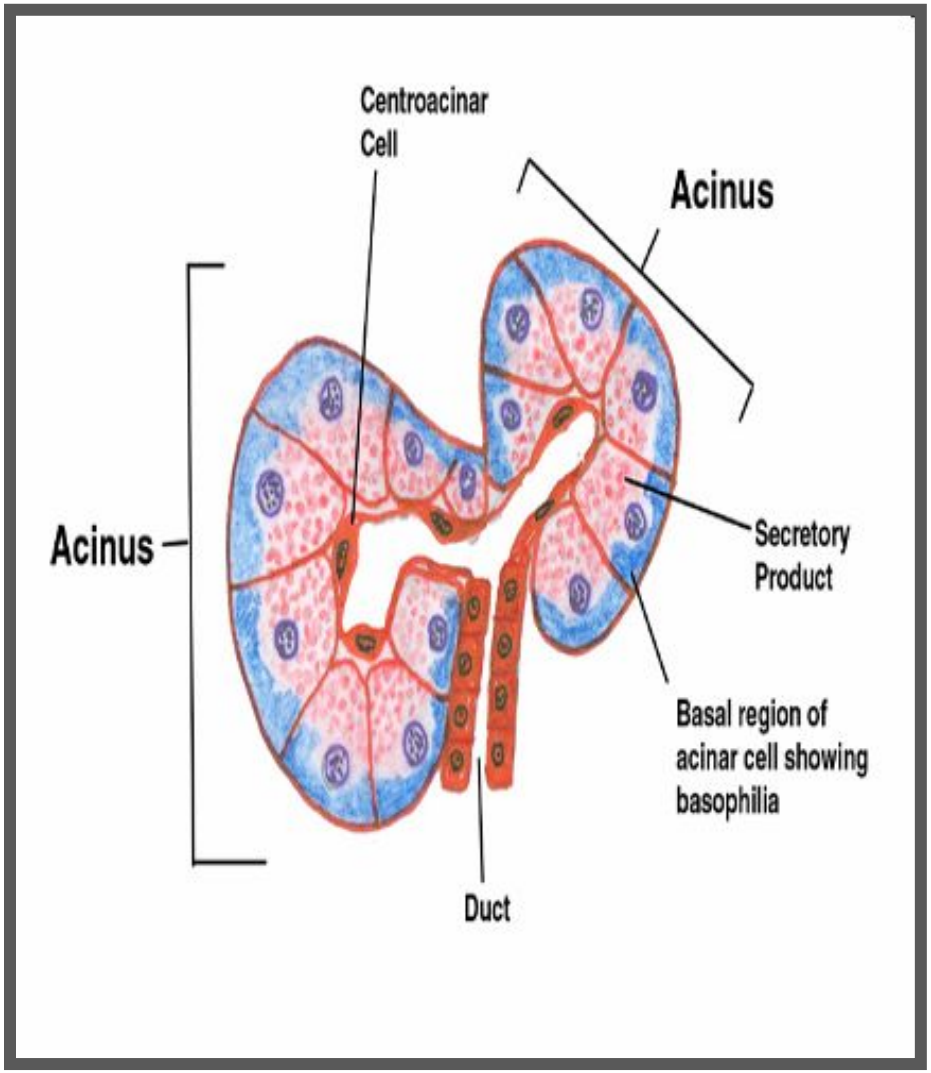
↓

Main Pancreatic Duct



ACINI & INTRALOBULAR DUCT



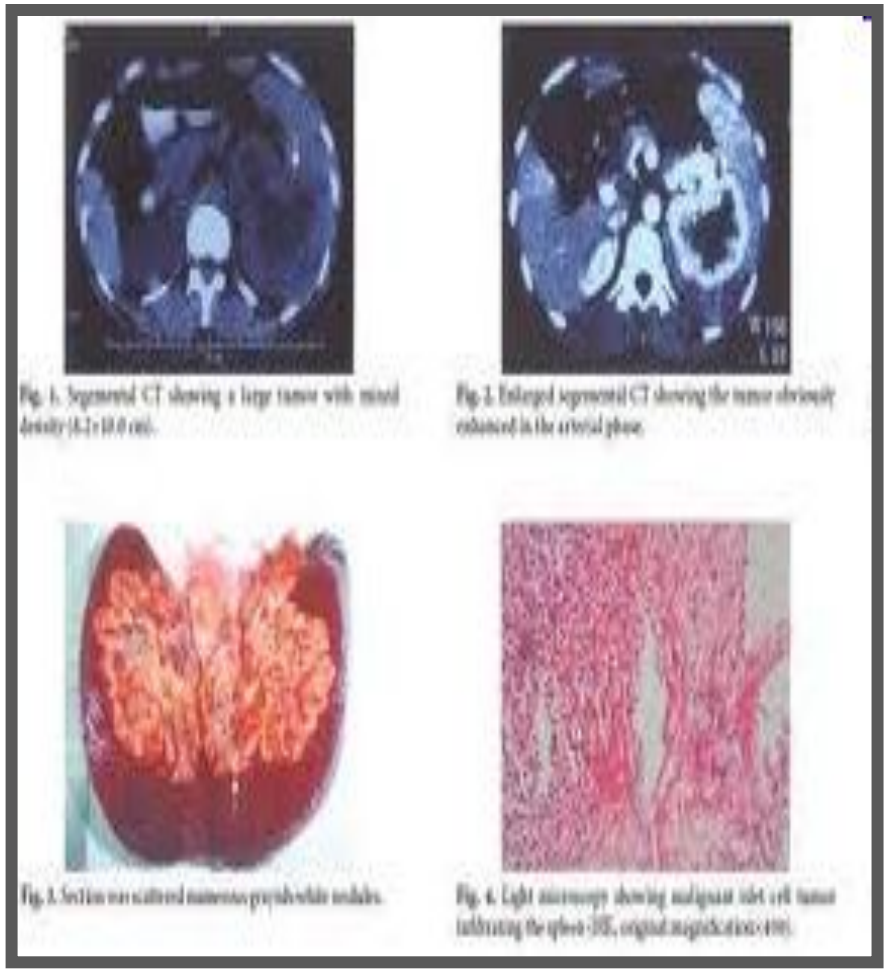


ACUTE PANCREATITIS

- Damage to the pancreatic acinar cells releases pancreatic enzymes into the local tissues.
- These powerful enzymes cause death of pancreatic tissue and severe inflammation termed *acute pancreatitis*.
- The release of pancreatic lipase causes death of local fat cells (*fat necrosis*).
- Pancreatic amylase is released and can be detected at high levels in the blood.
- This is a severe life-threatening condition.

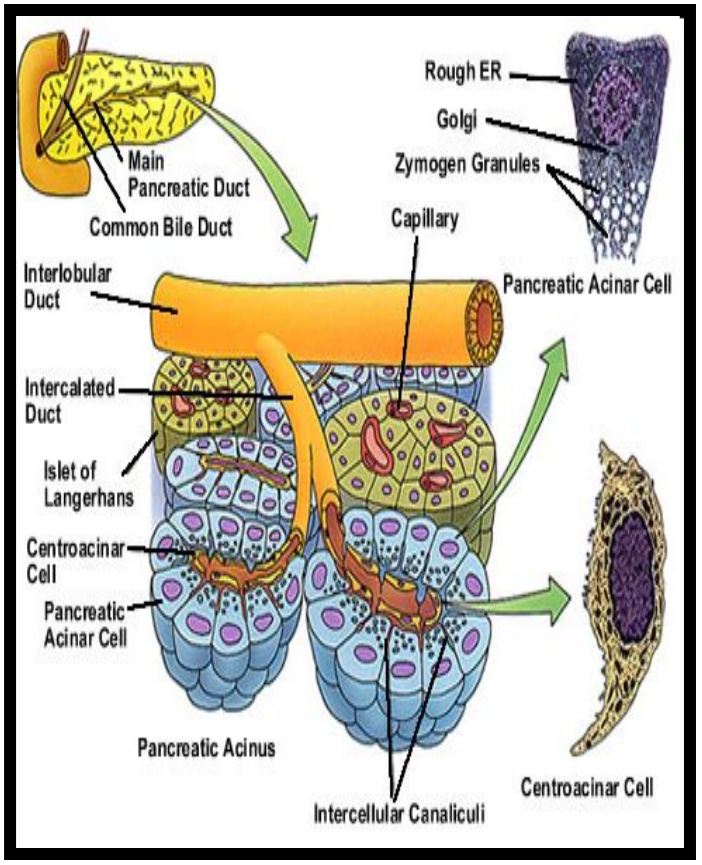
TUMORS OF ISLET CELLS

- Tumors of islet cells may produce insulin, glucagon, somatostatin, and pancreatic polypeptide.
- Some pancreatic tumors produce two or more of these hormones simultaneously, generating complex clinical symptoms.



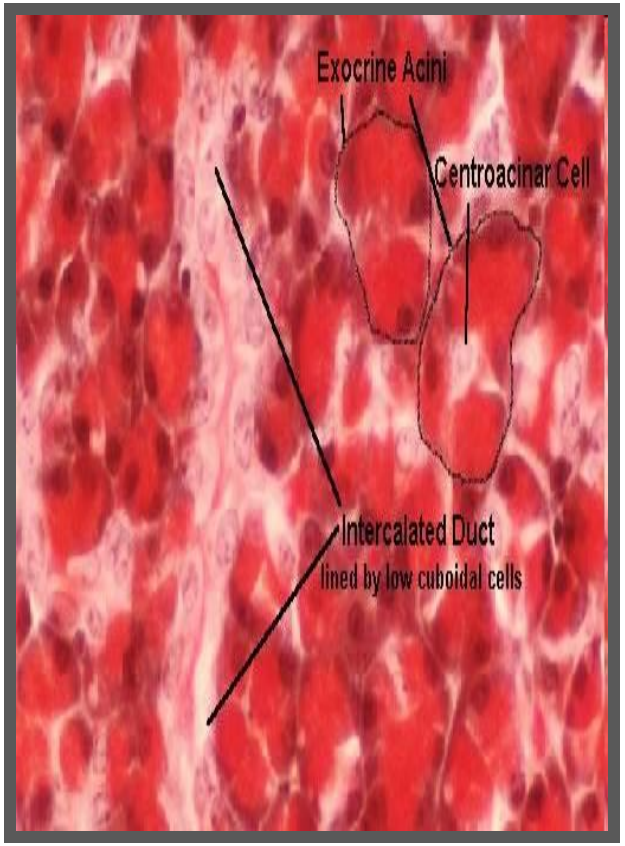
DIFFERENCES BETWEEN PANCREAS AND PAROTID GLAND

The exocrine portion of the pancreas is a compound tubulo-acinar gland, similar in structure to the parotid gland. In histological sections, a distinction between the two glands can be made based on the absence of **Striated Ducts** and the presence of the islets of Langerhans in the pancreas.



DIFFERENCES BETWEEN PANCREAS AND PAROTID GLAND

- Another characteristic detail is that in the pancreas the initial portions of intercalated ducts penetrate the lumens of the acini.
- Nuclei, surrounded by a pale cytoplasm, belong to **centroacinar cells** that constitute the intraacinar portion of the intercalated duct.
- These cells are found only in pancreatic acini.
- Intercalated ducts are tributaries of larger intralobular ducts that, in turn, form larger interlobular ducts lined by columnar epithelium, located within the connective tissue septa.
- There are no striated ducts in the pancreatic duct system.



THANK YOU